

Task No 02

Subject: Mobile App Development (MAD)

Section: BS SE 5-2

Submitted by:

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Submitted to:

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Task 1 – Splash Screen

```
import 'dart:async';
import 'dart:ui';
import 'package:flutter/material.dart';
import 'login_page.dart';

class WelcomeSplashScreen extends StatefulWidget {
  const WelcomeSplashScreen({super.key});

  State<WelcomeSplashScreen> createState() => _WelcomeSplashScreenState();
}

class _WelcomeSplashScreenState extends State<WelcomeSplashScreen>
    with SingleTickerProviderStateMixin {
  double _opacity = 1.0;
  double _blurValue = 0.0;
  late Timer _timer;

  void initState() {
    super.initState();

    _timer = Timer(const Duration(milliseconds: 3500), () {
      if (mounted) {
        setState(() {
          _opacity = 0.0;
          _blurValue = 15.0;
        });
      }
    });

    Future.delayed(const Duration(seconds: 5), () {
      if (mounted) {
        Navigator.of(context).pushReplacement(
          PageRouteBuilder(
            pageBuilder: (context, animation, secondaryAnimation) =>
                const LoginPage(),
            transitionsBuilder:
                (context, animation, secondaryAnimation, child) {
              return FadeTransition(opacity: animation, child: child);
```

```
        },
        transitionDuration: const Duration(milliseconds: 800),
      ),
    );
  }
 });
}

void dispose() {
 _timer.cancel();
 super.dispose();
}

Widget build(BuildContext context) {
 return Scaffold(
  body: Stack(
   children: [
    Container(
     decoration: const BoxDecoration(
      gradient: LinearGradient(
       colors: [Color(0xFF1A237E), Color(0xFF3F51B5), Color(0xFF5C6BC0)],
       begin: Alignment.topLeft,
       end: Alignment.bottomRight,
      ),
     ),
    ),
    AnimatedOpacity(
     duration: const Duration(milliseconds: 1500),
     opacity: _opacity,
     curve: Curves.easeInOut,
     child: TweenAnimationBuilder<double>(
      tween: Tween<double>(begin: 0.0, end: _blurValue),
      duration: const Duration(milliseconds: 1500),
      builder: (context, blur, child) {
       return ImageFiltered(
        imageFilter: ImageFilter.blur(sigmaX: blur, sigmaY: blur),
        child: child,
       );
      },
      child: Center(
```

```
child: Column(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    Container(
      padding: const EdgeInsets.all(22),
      decoration: const BoxDecoration(
        color: Colors.white24,
        shape: BoxShape.circle,
      ),
      child: const Icon(
        Icons.person_pin_rounded,
        size: 80,
        color: Colors.white,
      ),
    ),
    const SizedBox(height: 24),
    const Text(
      'Welcome to Wasi\'s Portfolio',
      style: TextStyle(
        fontSize: 30,
        fontWeight: FontWeight.bold,
        color: Colors.white,
        letterSpacing: 1.1,
      ),
    ),
    const SizedBox(height: 12),
    const Text(
      'Loading professional experience...',
      style: TextStyle(
        fontSize: 16,
        color: Colors.white70,
      ),
    ),
    const SizedBox(height: 40),
    const CircularProgressIndicator(
      color: Colors.white,
    ),
  ],
 ),
),
```

```
            ),
          ),
        ],
      ),
    );
  }
}
```

Task 2 – Login Page

```
import 'package:flutter/material.dart';

import 'package:flutter/services.dart';

import 'main.dart';


class LoginPage extends StatefulWidget {

  const LoginPage({super.key});


  @override

  State<LoginPage> createState() => _LoginPageState();

}


class _LoginPageState extends State<LoginPage> {

  final _usernameController = TextEditingController();

  final _passwordController = TextEditingController();

  final _formKey = GlobalKey<FormState>();


  bool _isPasswordVisible = false;

  bool _isLoading = false;

  String? _errorMessage;


  static const String _wasiUsername = 'wasi';

  static const String _wasiPassword = 'wasi123';
```

```
void _handleLogin() {
  FocusScope.of(context).unfocus();
  HapticFeedback.lightImpact();

  if (_formKey.currentState!.validate()) {
    setState(() {
      _isLoading = true;
      _errorMessage = null;
    });

    Future.delayed(const Duration(milliseconds: 800), () {
      if (!mounted) return;

      final username = _usernameController.text.trim();
      final password = _passwordController.text;

      if (username == _wasiUsername && password == _wasiPassword) {
        setState(() {
          _isLoading = false;
        });

        Navigator.of(context).pushReplacement(
          PageRouteBuilder(
            pageBuilder: (context, animation, secondaryAnimation) =>
                const HomePage(),
            transitionsBuilder:
                (context, animation, secondaryAnimation, child) {
              return FadeTransition(opacity: animation, child: child);
            },
```

```
            transitionDuration: const Duration(milliseconds: 600),
          ),
        );
      } else {
        setState(() {
          _isLoading = false;
          _errorMessage = 'Invalid Username or Password!';
        });
        HapticFeedback.vibrate();
      }
    });
  }
}

@override
void dispose() {
  _usernameController.dispose();
  _passwordController.dispose();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Container(
      decoration: const BoxDecoration(
        gradient: LinearGradient(
          colors: [Color(0xFFE8EAF6), Color(0xFFC5CAE9)],
          begin: Alignment.topCenter,
          end: Alignment.bottomCenter,
```

```
  ),
),
child: SafeArea(
  child: Center(
    child: SingleChildScrollView(
      padding: const EdgeInsets.symmetric(horizontal: 28.0, vertical: 20.0),
      child: Card(
        elevation: 6,
        shape: RoundedRectangleBorder(
          borderRadius: BorderRadius.circular(20),
        ),
        child: Padding(
          padding: const EdgeInsets.all(28.0),
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Container(
                padding: const EdgeInsets.all(16),
                decoration: BoxDecoration(
                  color: Colors.indigo.shade50,
                  shape: BoxShape.circle,
                ),
                child: const Icon(
                  Icons.verified_user_rounded,
                  size: 64,
                  color: Colors.indigo,
                ),
              ),
              const SizedBox(height: 20),
              const Text(
```

```
  'Wasi Portfolio Login',
  style: TextStyle(
    fontSize: 26,
    fontWeight: FontWeight.bold,
    color: Color(0xFF1A237E),
  ),
),
const SizedBox(height: 8),
const Text(
  'Sign in to access Wasi\'s profile',
  style: TextStyle(fontSize: 14, color: Colors.grey),
),
const SizedBox(height: 24),

Container(
  padding: const EdgeInsets.all(12),
  decoration: BoxDecoration(
    color: Colors.indigo.shade50,
    borderRadius: BorderRadius.circular(12),
    border: Border.all(color: Colors.indigo.shade200),
  ),
  child: const Row(
    children: [
      Icon(Icons.info_outline, color: Colors.indigo, size: 22),
      SizedBox(width: 10),
      Expanded(
        child: Text(
          'Login Credentials:\nUsername: wasi  |  Password: wasi123',
          style: TextStyle(
            fontSize: 13,
```

```
            fontWeight: FontWeight.w600,
            color: Color(0xFF1A237E),
          ),
        ),
      ),
    ],
  ),
),
const SizedBox(height: 24),

Form(
  key: _formKey,
  child: Column(
    children: [
      TextFormField(
        controller: _usernameController,
        decoration: InputDecoration(
          labelText: 'Username',
          hintText: 'wasi',
          prefixIcon: const Icon(Icons.person_outline),
          border: OutlineInputBorder(
            borderRadius: BorderRadius.circular(12),
          ),
        ),
        validator: (value) {
          if (value == null || value.trim().isEmpty) {
            return 'Please enter username';
          }
          return null;
        },
```

```
),
const SizedBox(height: 16),

TextFormField(
  controller: _passwordController,
  obscureText: !_isPasswordVisible,
  decoration: InputDecoration(
    labelText: 'Password',
    hintText: 'wasi123',
    prefixIcon: const Icon(Icons.lock_outline),
    suffixIcon: IconButton(
      icon: Icon(
        _isPasswordVisible
            ? Icons.visibility_off
            : Icons.visibility,
      ),
      onPressed: () {
        setState(() {
          _isPasswordVisible = !_isPasswordVisible;
        });
      },
    ),
    border: OutlineInputBorder(
      borderRadius: BorderRadius.circular(12),
    ),
  ),
  validator: (value) {
    if (value == null || value.isEmpty) {
      return 'Please enter password';
    }
```

```
    return null;
  },
),
const SizedBox(height: 12),

if (_errorMessage != null)
  Padding(
    padding: const EdgeInsets.only(bottom: 12.0),
    child: Text(
      _errorMessage!,
      style: const TextStyle(
        color: Colors.red,
        fontWeight: FontWeight.bold,
        fontSize: 14,
      ),
    ),
  ),

const SizedBox(height: 12),

SizedBox(
  width: double.infinity,
  height: 50,
  child: ElevatedButton(
    onPressed: _isLoading ? null : _handleLogin,
    style: ElevatedButton.styleFrom(
      backgroundColor: const Color(0xFF1A237E),
      foregroundColor: Colors.white,
      shape: RoundedRectangleBorder(
        borderRadius: BorderRadius.circular(12),
```

```
                ),
                elevation: 4,
              ),
              child: _isLoading
                  ? const SizedBox(
                      width: 24,
                      height: 24,
                      child: CircularProgressIndicator(
                        color: Colors.white,
                        strokeWidth: 2.5,
                      ),
                    )
                  : const Text(
                      'LOGIN',
                      style: TextStyle(
                        fontSize: 16,
                        fontWeight: FontWeight.bold,
                      ),
                    ),
            ),
          ),
        ),
      ],
    ),
  ),
),
```

```
        ),
      ),
    );
  }
}
```

Task 3 – Animated Widget

```
import 'package:flutter/material.dart';


class AnimatedContainerWidget extends StatefulWidget {
  const AnimatedContainerWidget({super.key});


  @override
  State<AnimatedContainerWidget> createState() =>
      _AnimatedContainerWidgetState();
}


class _AnimatedContainerWidgetState
    extends State<AnimatedContainerWidget> {
  double _height = 100;
  double _width = 100;
  Color _color = Colors.blue;
  double _borderRadius = 12.0;


  void _toggleContainer() {
    setState(() {
      if (_width == 100) {
        _height = 180;
        _width = 180;
```

```
      _color = Colors.deepOrangeAccent;
      _borderRadius = 30.0;
    } else {
      _height = 100;
      _width = 100;
      _color = Colors.blue;
      _borderRadius = 12.0;
    }
  });
}

@override
Widget build(BuildContext context) {
  return GestureDetector(
    onTap: _toggleContainer,
    child: AnimatedContainer(
      duration: const Duration(milliseconds: 600),
      curve: Curves.easeInOutBack,
      height: _height,
      width: _width,
      decoration: BoxDecoration(
        color: _color,
        borderRadius: BorderRadius.circular(_borderRadius),
        boxShadow: [
          BoxShadow(
            color: _color.withValues(alpha: 0.3),
            blurRadius: 10,
            offset: const Offset(0, 4),
          ),
        ],
```

```
        ),
        child: const Center(
          child: Text(
            'Tap Box\nTo Transform',
            textAlign: TextAlign.center,
            style: TextStyle(
              color: Colors.white,
              fontWeight: FontWeight.bold,
              fontSize: 14,
            ),
          ),
        ),
      ),
    ),
  );
 }
}
```

Task 4 – Animated Buttons

```
import 'package:flutter/material.dart';

class AnimatedCrossFadeWidget extends StatefulWidget {
  const AnimatedCrossFadeWidget({super.key});

  @override
  State<AnimatedCrossFadeWidget> createState() => AnimatedCrossFadeWidgetState();
}

class AnimatedCrossFadeWidgetState extends State<AnimatedCrossFadeWidget> {
  bool _crossFadeStateShowFirst = true;
```

```
void _crossFade() {
  setState(() {
    _crossFadeStateShowFirst = !_crossFadeStateShowFirst;
  });
}

@override
Widget build(BuildContext context) {
  return Row(
    mainAxisAlignment: MainAxisAlignment.center,
    children: <Widget>[
      Stack(
        alignment: Alignment.center,
        children: <Widget>[
          AnimatedCrossFade(
            duration: const Duration(milliseconds: 500),
            sizeCurve: Curves.bounceOut,
            crossFadeState: _crossFadeStateShowFirst
                ? CrossFadeState.showFirst
                : CrossFadeState.showSecond,
            firstChild: Container(
              color: Colors.amber,
              height: 100.0,
              width: 100.0,
            ),
            secondChild: Container(
              color: Colors.lime,
              height: 300.0,
              width: 300.0,
```

```
            ),
          ),
          Positioned.fill(
            child: FloatingActionButton(
              onPressed: _crossFade,
              child: const Text('Tap to\nFade', textAlign: TextAlign.center),
            ),
          ),
        ],
      ),
    ],
  );
 }
}
```

Task 5 – Main

```
export 'src/week_08_base.dart';


import 'package:flutter/material.dart';
import 'package:flutter/services.dart';
import 'animated_container_widget.dart';
import 'splash_screen.dart';
import 'login_page.dart';


void main() {
 runApp(const MyApp());
}


class MyApp extends StatelessWidget {
```

```
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Wasi - Profile & Portfolio',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.indigo),
        useMaterial3: true,
      ),
      home: const WelcomeSplashScreen(),
    );
  }
}

class HomePage extends StatefulWidget {
  const HomePage({super.key});

  @override
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
  bool _isExpanded = false;
  bool _showWelcome = true;
  double _fabScale = 1.0;

  void _onPlusPressed() {
    HapticFeedback.lightImpact();
```

```
  setState(() {
    _isExpanded = !_isExpanded;
    _showWelcome = true;
    _fabScale = 0.85;
  });

  Future.delayed(const Duration(milliseconds: 150), () {
    if (mounted) {
      setState(() {
        _fabScale = 1.0;
      });
    }
  });
}

void _handleLogout() {
  HapticFeedback.lightImpact();
  Navigator.of(context).pushReplacement(
    PageRouteBuilder(
      pageBuilder: (context, animation, secondaryAnimation) =>
          const LoginPage(),
      transitionsBuilder: (context, animation, secondaryAnimation, child) {
        return FadeTransition(opacity: animation, child: child);
      },
      transitionDuration: const Duration(milliseconds: 600),
    ),
  );
}
```

```
void _showContactDialog() {
  showDialog(
    context: context,
    builder: (BuildContext context) {
      return AlertDialog(
        shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(16)),
        title: const Text('Wasi\'s Contact Details'),
        content: Column(
          mainAxisSize: MainAxisSize.min,
          crossAxisAlignment: CrossAxisAlignment.start,
          children: const [
            Text('📱 Phone: 0300 7654321'),
            SizedBox(height: 8),
            Text('📧 Email: wasi@example.com'),
            SizedBox(height: 8),
            Text('📍 Location: Pakistan'),
          ],
        ),
        actions: [
          TextButton(
            onPressed: () => Navigator.of(context).pop(),
            child: const Text('Close'),
          ),
        ],
      );
    },
  );
}
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Wasi - Profile & Portfolio'),
      centerTitle: true,
      elevation: 2,
      backgroundColor: Colors.indigo.shade800,
      foregroundColor: Colors.white,
      actions: [
        IconButton(
          icon: const Icon(Icons.logout),
          tooltip: 'Logout',
          onPressed: _handleLogout,
        ),
      ],
    ),
    body: SafeArea(
      child: Center(
        child: SingleChildScrollView(
          padding: const EdgeInsets.all(24.0),
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            crossAxisAlignment: CrossAxisAlignment.center,
            children: <Widget>[
              // Welcome Banner
              AnimatedSwitcher(
                duration: const Duration(milliseconds: 600),
                transitionBuilder: (Widget child, Animation<double> animation) {
                  return FadeTransition(
```

```
    opacity: animation,
    child: SizeTransition(
      sizeFactor: animation,
      child: child,
    ),
  );
},
child: _showWelcome
    ? Container(
        key: const ValueKey('welcome_banner'),
        margin: const EdgeInsets.only(bottom: 24),
        padding: const EdgeInsets.all(16),
        decoration: BoxDecoration(
          color: Colors.indigo.shade50,
          borderRadius: BorderRadius.circular(16),
          border: Border.all(color: Colors.indigo.shade200),
          boxShadow: [
            BoxShadow(
              color: Colors.indigo.withValues(alpha: 0.1),
              blurRadius: 10,
              offset: const Offset(0, 4),
            ),
          ],
        ),
        child: Row(
          children: [
            const Icon(Icons.verified_user, color: Colors.indigo, size: 28),
            const SizedBox(width: 12),
            Expanded(
              child: Text(
```

```
                _isExpanded
                    ? 'Welcome Wasi! Profile expanded & interactive animations active.'
                    : 'Welcome back, Wasi! Tap the plus button to animate your profile.',
                style: TextStyle(
                  fontSize: 15,
                  fontWeight: FontWeight.w600,
                  color: Colors.indigo.shade900,
                ),
              ),
            ),
          ],
        ),
      )
    : const SizedBox.shrink(key: ValueKey('empty_banner')),
),

// Interactive Profile Picture with smooth zoom animation (wasi.jpeg)
Semantics(
  label: 'Wasi profile picture, animates in size when interactive button is pressed',
  child: AnimatedContainer(
    duration: const Duration(milliseconds: 600),
    curve: Curves.elasticOut,
    width: _isExpanded ? 160 : 110,
    height: _isExpanded ? 160 : 110,
    decoration: BoxDecoration(
      shape: BoxShape.circle,
      border: Border.all(color: Colors.indigo, width: 3),
      boxShadow: [
        BoxShadow(
          color: Colors.black.withValues(alpha: 0.2),
```

```
            blurRadius: _isExpanded ? 18 : 8,
            offset: const Offset(0, 6),
          ),
        ],
      ),
      child: CircleAvatar(
        radius: _isExpanded ? 80 : 55,
        backgroundImage: const AssetImage('assets/images/wasi.jpeg'),
      ),
    ),
  ),
  const SizedBox(height: 20),

  const Text(
    'Wasi',
    style: TextStyle(
      fontSize: 30,
      fontWeight: FontWeight.bold,
      color: Colors.indigo,
    ),
  ),
  const SizedBox(height: 6),
  const Text(
    'BS Software Engineering Student & Flutter Developer',
    textAlign: TextAlign.center,
    style: TextStyle(
      fontSize: 16,
      color: Colors.grey,
      fontWeight: FontWeight.w500,
    ),
```

```
            ),
            const SizedBox(height: 16),

            const Icon(
              Icons.person,
              size: 32,
              color: Colors.indigo,
            ),
            const SizedBox(height: 16),

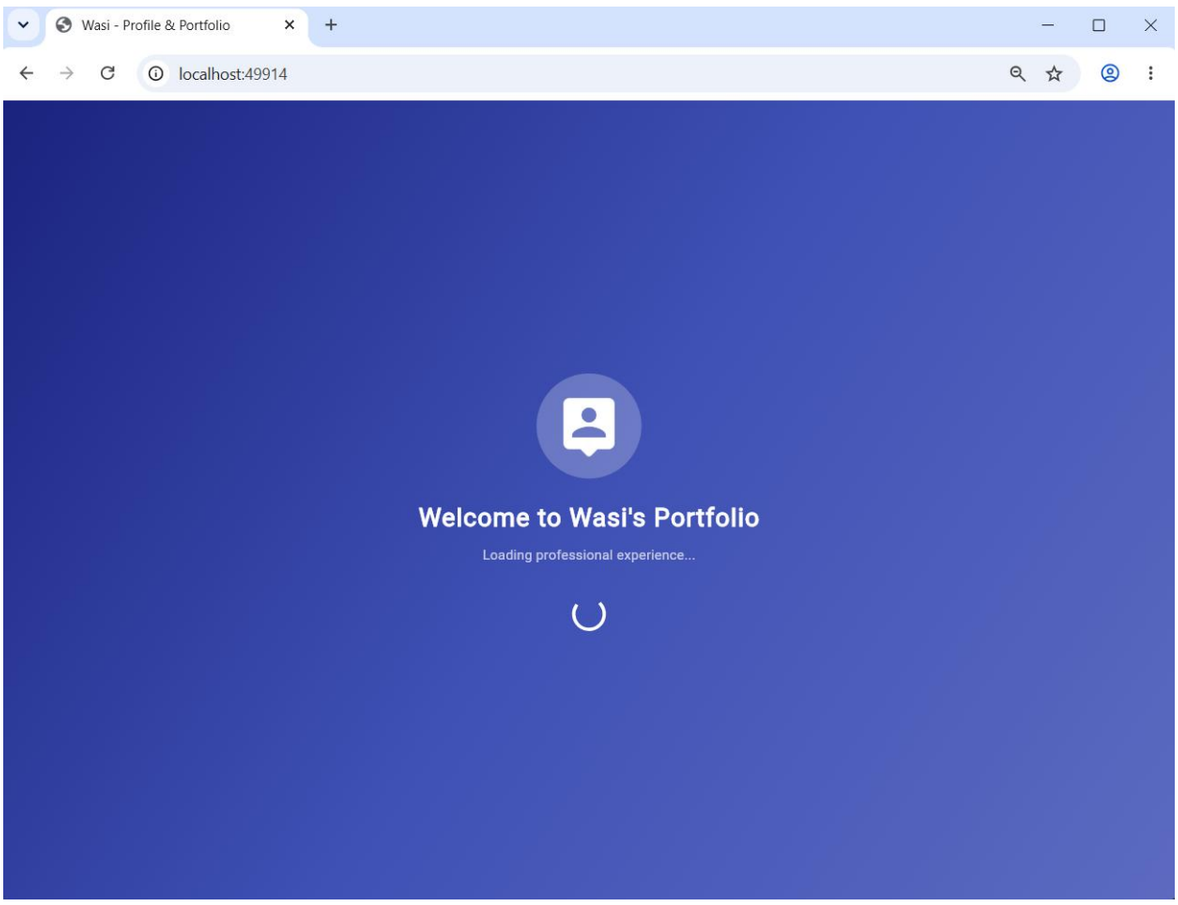
            // Bio Card
            Card(
              elevation: 2,
              shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(16)),
              child: Padding(
                padding: const EdgeInsets.all(20),
                child: const Text(
                  'Hello! I am Wasi. I am a student of BS Software Engineering and a passionate
Flutter Developer. '
                  'I love building intuitive, beautiful cross-platform mobile apps with modern
UI/UX, responsive design, and smooth animations.',
                  textAlign: TextAlign.center,
                  style: TextStyle(
                    fontSize: 16,
                    height: 1.4,
                  ),
                ),
              ),
            ),
            const SizedBox(height: 24),
```

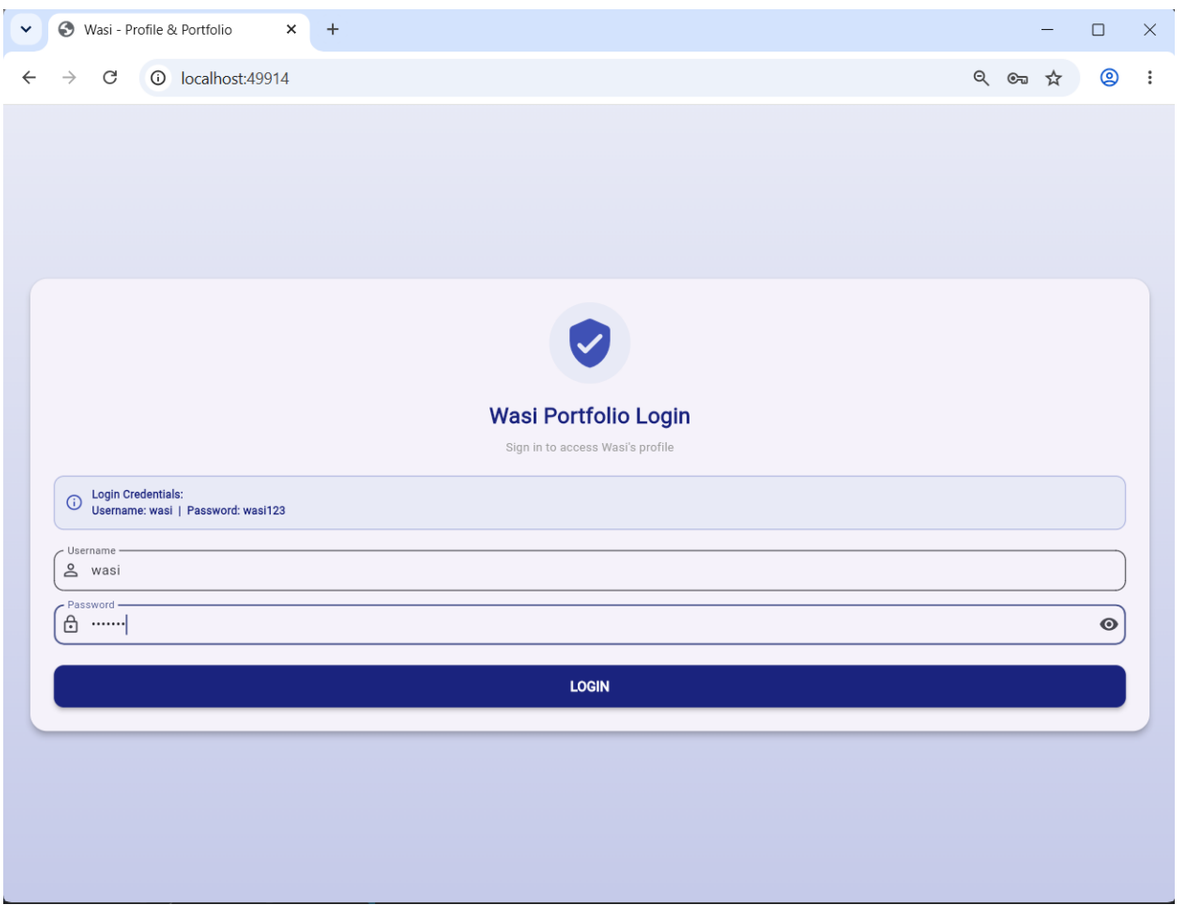
```
          // Contact Me Button
          ElevatedButton.icon(
            onPressed: _showContactDialog,
            icon: const Icon(Icons.contact_mail),
            label: const Text('Contact Me'),
            style: ElevatedButton.styleFrom(
              backgroundColor: Colors.indigo.shade800,
              foregroundColor: Colors.white,
              padding: const EdgeInsets.symmetric(horizontal: 24, vertical: 12),
              shape: RoundedRectangleBorder(
                borderRadius: BorderRadius.circular(12),
              ),
            ),
          ),
          const SizedBox(height: 30),

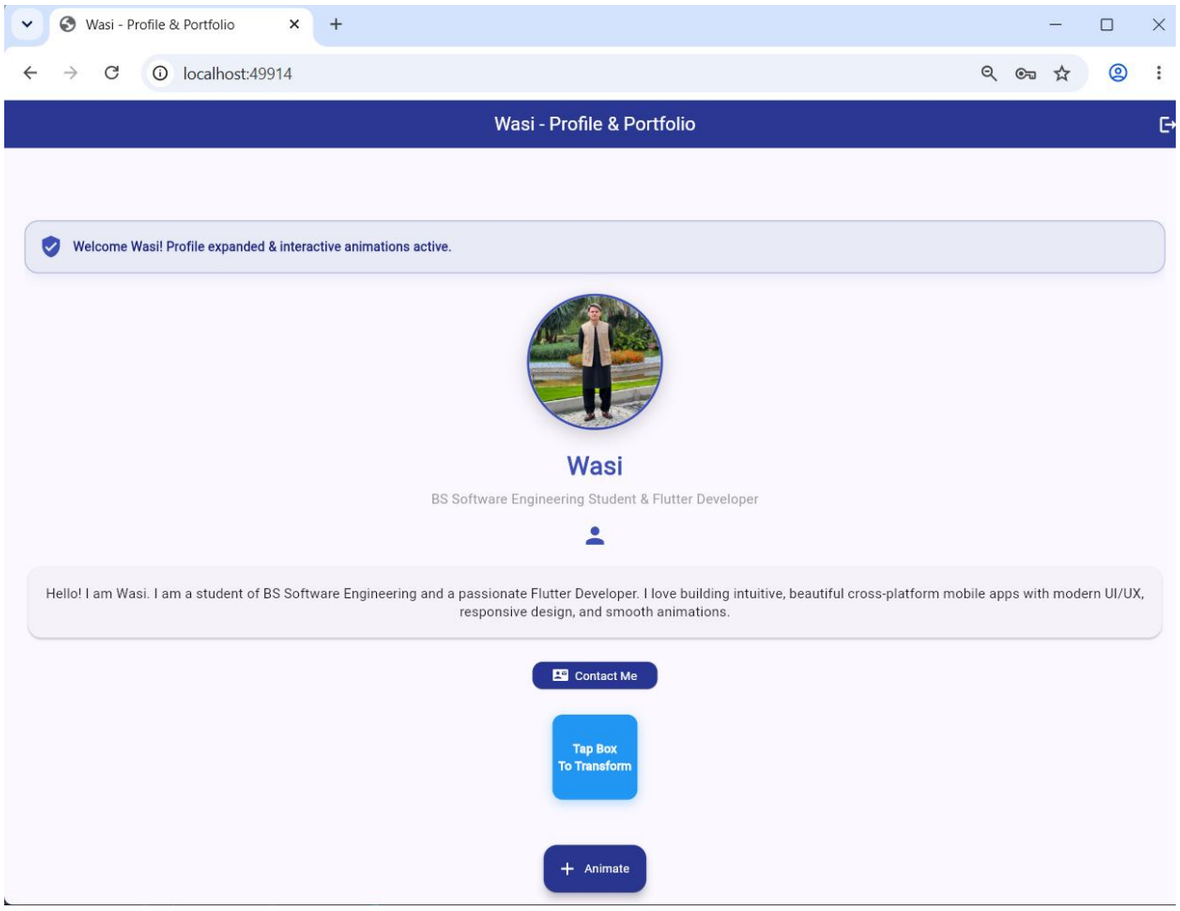
          // Animated Container Widget from custom module
          const AnimatedContainerWidget(),
          const SizedBox(height: 40),
        ],
      ),
    ),
  ),
),
floatingActionButton: Semantics(
  label: 'Interactive plus button to trigger animations',
  child: AnimatedScale(
    scale: _fabScale,
    duration: const Duration(milliseconds: 150),
```

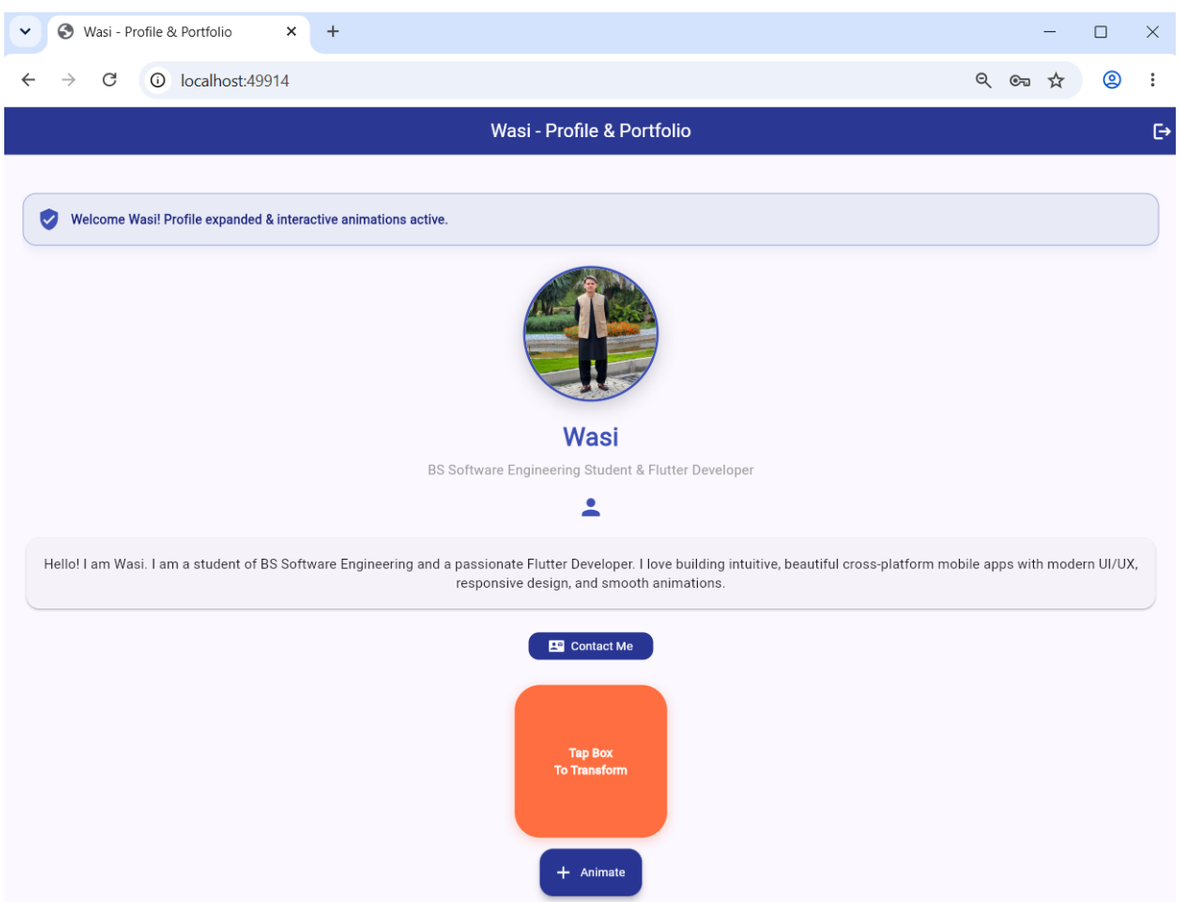
```
          curve: Curves.easeInOut,
          child: FloatingActionButton.extended(
            onPressed: _onPlusPressed,
            icon: const Icon(Icons.add),
            label: const Text('Animate'),
            backgroundColor: Colors.indigo.shade800,
            foregroundColor: Colors.white,
            elevation: 6,
          ),
        ),
      ),
      floatingActionButtonLocation: FloatingActionButtonLocation.centerFloat,
    );
  }
}
```

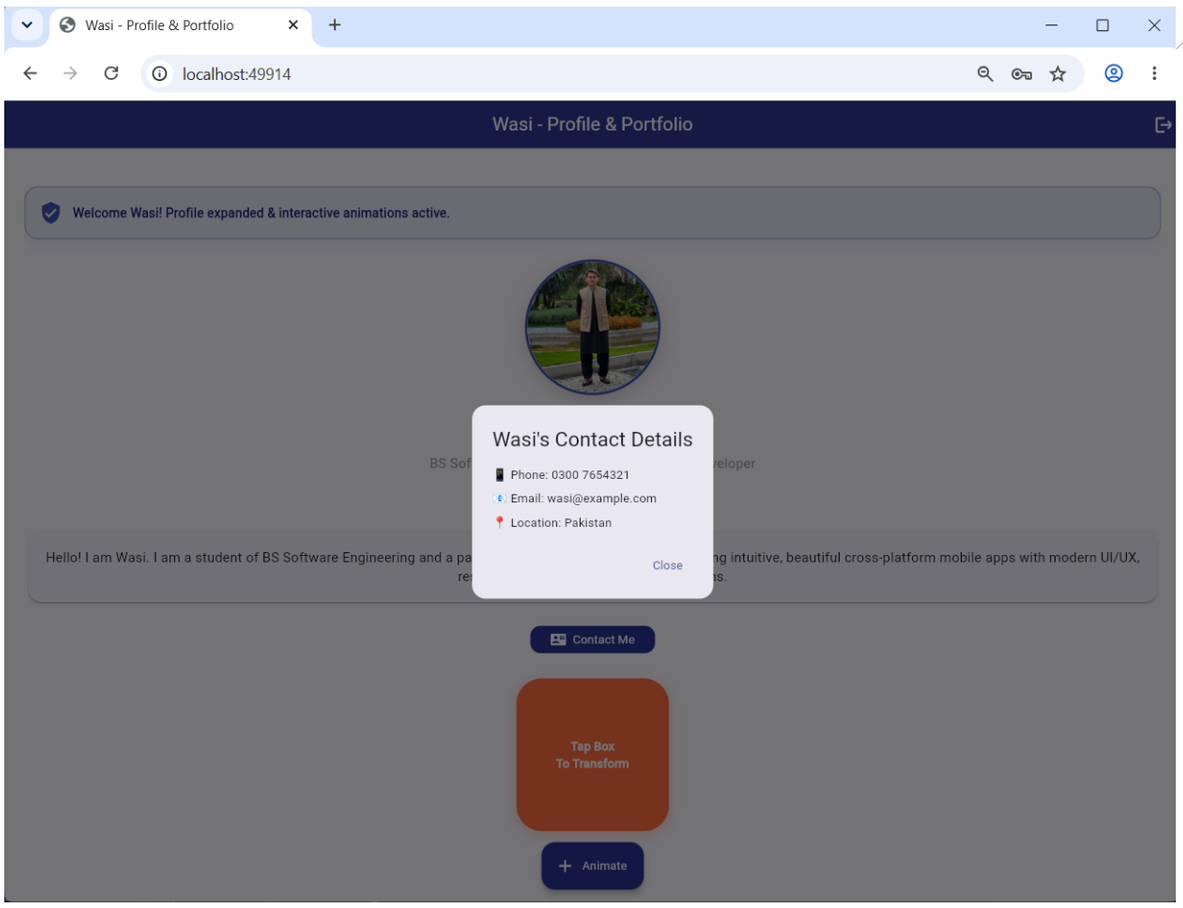
Outputs Screen Shots:






Wasi - Profile & Portfolio
localhost:49914
Wasi - Profile & Portfolio
Welcome Wasi! Profile expanded & interactive animations active.
Wasi
BS Software Engineering Student & Flutter Developer
Hello! I am Wasi. I am a student of BS Software Engineering and a passionate Flutter Developer. I love building intuitive, beautiful cross-platform mobile apps with modern UI/UX, responsive design, and smooth animations.
Contact Me
Tap Box To Transform
Animate


Wasi - Profile & Portfolio
localhost:49914
Wasi - Profile & Portfolio
Welcome Wasi! Profile expanded & interactive animations active.
Wasi
BS Software Engineering Student & Flutter Developer
Hello! I am Wasi. I am a student of BS Software Engineering and a passionate Flutter Developer. I love building intuitive, beautiful cross-platform mobile apps with modern UI/UX, responsive design, and smooth animations.
Contact Me
Tap Box To Transform
Animate


Wasi - Profile & Portfolio
localhost:49914
Wasi - Profile & Portfolio
Welcome Wasi! Profile expanded & interactive animations active.
Wasi's Contact Details
Phone: 0300 7654321
Email: wasi@example.com
Location: Pakistan
Close
Contact Me
Tap Box
To Transform
Animate